

Scalability & Future Plan

Date	4 July 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	1 Mark

Scalability & Future Plan:

The **Voice-Based Concept Understanding Analyser (VBCUA)** is designed with a modular architecture that supports future enhancements and scalability. While the current implementation successfully evaluates spoken conceptual explanations for individual users, the system can be extended to support multiple users, cloud deployment, database integration, and advanced AI capabilities.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local file storage is used instead of a centralized database.	Limits data persistence and multi-user access.	High
2	Supports only single-user execution on a local Streamlit application.	Cannot efficiently handle concurrent users.	High
3	User authentication and evaluation history are not implemented.	Users cannot securely access or track previous assessments.	medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a single/local user.	Deploy on a cloud platform with support for multiple concurrent users using FastAPI and load balancing.
2	Data Storage	Audio files and reports are stored locally.	Integrate a PostgreSQL/SQLite database and cloud storage (e.g., AWS S3 or Google Cloud Storage) for scalable data management.
3	Performance	AI models execute sequentially on a local machine.	Optimize model loading, enable GPU acceleration, and implement asynchronous/background processing to reduce response time.
4	Security	Basic local application with no authentication.	Add secure user authentication (JWT/OAuth2), encrypted data storage, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	User authentication, database integration, evaluation history, and user profiles.	3–6 Months	Enables secure multi-user access and long-term progress tracking.
Phase 2	Cloud deployment, REST API development, multilingual speech support, and real-time voice recording.	6–12 Months	Improves accessibility, scalability, and supports a wider range of users.
Phase 3	Personalized learning recommendations, AI chatbot assistance, advanced analytics dashboard, and mobile application support.	12+ Months	Enhances user experience, educational insights, and expands the platform for large-scale educational institutions.